ALLIED WELDING SYSTEMS

How It Works

A plain guide to the Operations Bridge: what it does to an order, what it saves, what it costs, and what it takes to keep it running.

The whole thing, in one paragraph

Every order Allied takes absorbs about two hours of someone's day in typing, looking up and chasing. This system does that work in a twenty-fifth of a second and hands your team a five-minute review instead. On the way through, it checks every supplier bill against what you actually authorised — which nobody does today — and refuses to proceed when something is wrong. It costs $6,000 once, runs on a computer you already own, and sends none of your pricing outside the building.

The four numbers

2 hrs **becomes 5 minutes.** Per order. At twenty orders a week that is **1,993 hours a year** — a full-time person handed back, as capacity rather than redundancy.

10 wks **to pay for itself** at ten orders a week. Five weeks at twenty. The decision is reversible well inside a quarter.

Every **supplier bill gets checked**, line by line, freight included, against the order you raised. That does not happen today, and it is where the money actually is — time is just the part that is easy to count.

94% **of the busy work can go.** Plugging it in removes 88% — supplier bills are now checked the moment they are read. Two more small pieces of plumbing, about a week, take it to 94%.

> **One thing to know before you read on**
>
> The system is **built and tested, but not yet plugged in**. Today it produces the Sage files correctly but someone still imports them, and it records what it would post to your Monday board rather than posting it. Connecting both is a few days' work and is covered in the separate proposal. Everything else in this guide works now.

1. The problem

Every order that comes in gets touched by a person four or five times. Someone reads the customer's email. Someone types it into Sage. Someone decides which supplier gets what and raises a purchase order for each. Someone checks the supplier's bill when it arrives, or more honestly, glances at it. Someone builds the packing slip and finds the mill certs.

That is about an afternoon per order, and it is the same afternoon whether the order is nine hundred dollars or ninety thousand.

> **Why that matters:** It means your office, not your shop floor, decides how much work Allied can take. You can always find another fabricator. You cannot always find another afternoon.

The system removes the typing and keeps the judgement. It reads the order, works out what is being bought and from whom, raises the paperwork, and checks the supplier's invoice when it lands. Your team reviews instead of transcribes.

2. Where the money actually is

Four places. They are worth very different amounts, and the order below is deliberate.

Supplier invoices you are currently paying without checking

When a supplier bills you, the system compares every line against the purchase order you raised — the price, the quantity, and the freight. If Nibrock quoted $110 a unit and bills $128.50, it flags the difference and writes the dispute letter for you.

It also catches two things a busy person almost never does:

- A bill quoting a purchase order **you never raised at all**. The system refuses it and holds the payment.
- A line for something **that was never on the order**. A $50,000 charge for goods you did not buy shows as a $50,000 problem, not a rounding note.

> **Why that matters:** You are a brokerage. You do not make money on the steel — you make it on the spread between what you buy at and what you sell at. So every dollar a supplier overbills comes straight out of the only thing you actually sell. A few percent of leakage on the buy side is a few percent off the whole business, not off the invoice.

Order volume without more office staff

Per-order admin drops from an afternoon to a few minutes of review. This is not about replacing anyone — it is about the ceiling moving.

> **Why that matters:** Today, taking on 30% more work means hiring in the office before you see a dollar of it. That is a decision with a payroll attached, so it gets deferred, and the growth gets deferred with it.

Quoting off steel prices that have moved

If a supplier's price is more than 60 days old, the system stops the order before it commits, asks the supplier for a fresh number, and puts the job in a review list.

> **Why that matters:** The damage from a stale price is not the price — it is that you have already told the customer. Once a number is quoted, the choice is eating the difference or going back on it. The system catches it while it is still just a number on a screen.

Paperwork that cannot embarrass you

Blind shipping is enforced automatically. The packing slip in the crate carries Allied's name, never the supplier's, and never shows your cost. Mill certs are matched to the actual heat numbers on the order.

> **Why that matters:** One slip here does not cost you an order — it costs you the customer, because you have just shown them who they could buy from directly.

3. What it saves, in hours

This is the part that is easy to measure, so it is worth doing properly rather than waving at.

The same order, both ways

The work	By hand today	With the system
Read the order and key it into Sage	20–30 min	done for you
Work out which supplier gets what, raise a PO each	25–40 min	done for you
Build the packing slip and dig out the mill certs	30–45 min	done for you
Check the supplier bill line by line	15–25 min, usually skipped	done for you
Update the board and the file	10 min	done for you
Someone reads it and approves	—	**about 5 min**
Total	**about 2 hours**	**about 5 minutes**

The right-hand column is measured. The left-hand column is the one figure in this document we are taking on trust — it is an estimate of Allied's own hand-work, and you are better placed to correct it than we are. Every number below moves with it.

The machine time, measured

A whole order — read, matched to your catalogue, split between suppliers, run through every check, sales order and purchase orders written, packing slip produced — takes **25 thousandths of a second**. A full line-by-line check of a supplier bill, including drafting the dispute letter, takes **8 thousandths**. The system can do about forty complete orders in the time it takes you to blink. **Nothing you do will ever be the thing it is waiting on.**

Why that matters: It matters that the machine side is effectively instant, because it means the only thing that limits how much work Allied can take is how fast people decide — not how fast anyone types. That is the whole point. You have moved the bottleneck from the keyboard to the judgement, and judgement is the part you were actually hiring for.

What that is worth in a year

Read your own row. The only assumption is the two hours above; everything else is arithmetic at **$30 an hour** fully loaded.

Orders a week	Orders a year	Hours back	Worth	Pays back the $6,000 in
5	260	498 h	$14,950	21 weeks
10	520	997 h	$29,900	10 weeks
20	1,040	1,993 h	$59,800	5 weeks
40	2,080	3,987 h	$119,600	3 weeks

At twenty orders a week you get a full-time person back

Not as a redundancy — as capacity. Around **1,993 hours a year** stop being spent on typing. That is one person's entire working year returned to quoting, chasing suppliers and answering customers, which is work that brings money in rather than work that merely records it.

Why that matters: The reason to care about the payback period rather than the yearly figure is that it tells you what you are risking. At ten orders a week the system has paid for itself before the end of the quarter you bought it in. If it then disappointed you entirely, you would be out one quarter of one person's admin time — which is roughly the cost of thinking about it for too long.

4. The two systems it sits between

Monday.com — your shared board

You already use it. Think of it as a very large whiteboard where each row is a job and the columns say which customer, what stage, who owns it. Its weakness today is that it is only as current as the last person who remembered to update it.

The system works with it both ways:

- **The board tells the system when something happens.** Drag a job to “Shipped” and the packing list and dispatch paperwork are produced automatically.
- **The system writes back to the board.** Order numbers, supplier PO numbers and status appear on the card without anyone typing them.

> **Why that matters:** Nothing changes about how your team uses the board. They keep dragging cards. The difference is the card fills itself in, and dragging it now actually does something.

Sage 50 — your books

Sage is the official record of what was sold, bought, owed and owned. What it does not do is work anything out. It records what it is told. It does not read the customer's email, does not know which of three suppliers should get the plate, and does not check Nibrock's bill against your order.

The system sits in front of Sage and does the thinking. Sage stays the book of record. Nothing about your accounting changes; the entries simply arrive correct and untyped.

Because you buy and sell on every job, each order is really two transactions, and both have to reach Sage:

Document	Direction	Records
Sales Order	Money in	What the customer bought, at your price
Purchase Order	Money out	What you bought from each supplier, at their price

One customer order can produce one sales order and three purchase orders, because the plate comes from Nibrock, the screens from KWS and the hose from Zibo.

> **Why that matters:** Keeping those two sides tied together is what makes the invoice checking possible at all. Without the link, a supplier bill is just a number with nothing to compare it to — which is the situation today.

> **One honest wrinkle about Sage**
>
> Sage 50 has no built-in way for outside software to write into it directly unless you run the **desktop** version on the same machine. If you are on the newer online Sage, the system produces the import files and someone clicks import — which still removes all the typing, just not the last click. We check which you have before committing to anything. It takes under a minute.

5. What happens to an order

	Step	Replaces
1	Reads the order out of an email or PDF — the PO number, the customer, the lines	Reading and re-keying
2	Matches each line to your catalogue, even when the customer uses their own part name	Looking up part numbers
3	Splits the order by supplier, or marks it for your own stock	Deciding who supplies what
4	**Runs the checks.** If one fails, the order stops and goes to a review list	**Nothing — this did not happen before**
5	Creates the sales order and one purchase order per supplier	Typing into Sage
6	Produces the packing slip, mill certs, and customs papers if it crosses a border	Assembling documents by hand
7	Updates the Monday board	Remembering to update the board

> **Why that matters:** Steps 1, 2, 3, 5, 6 and 7 save time, and time is cheap to value. Step 4 is the one you are actually buying, because it is the only one that was not happening at all.

What the system refuses to do

It will not	Which prevents
Approve a bill citing an order you never raised	Paying for a job that was never yours
Let a charge for unordered goods cancel itself out	A large overcharge reading as zero
Quote off a supplier price over 60 days old	Committing to a number steel has moved past
Pass an order below 15% gross margin unflagged	Winning work that loses money
Raise a second set of POs for a repeated order	Double-ordering from a resent email
Certify material with no mill cert on file	Signing for something you cannot prove
Certify a mill report with missing readings	Treating 'not measured' as 'passed'

It will not	Which prevents
Put a supplier's name on a customer packing slip	Handing over your supply chain

How you know these hold

Each one has an automatic check that fails the moment anybody breaks it. There are 150 such checks, 60 full order run-throughs, and 27 deliberate attempts to sneak a bad invoice or an uncertifiable heat past the system. All pass, and they run in under ten seconds.

Why that matters: This is the difference between a promise and a guarantee. Anyone can say the rules are enforced. These rules are written down in a form that complains loudly if a future change breaks one — including a change made by someone who has never met you.

6. How much further this can go

Plugging it in removes most of the busy work, but not all. It is worth being precise about what is left, because the remainder is cheap to remove and nobody should have to guess at it.

Minutes of hand-work left on each order

Stage	Left by hand	Busy work gone	Hours back a year at 20 orders a week
Today, by hand	120 min	0%	—
Once it is plugged in (this proposal)	14 min	88%	1,837 h
Plus collecting mail and sending it	7 min	94%	1,959 h
Plus writing straight into Sage	5 min	96%	1,993 h

What those 14 minutes actually are

Someone forwards the order email (2 min), someone clicks import in Sage (2 min), someone emails the purchase orders to the suppliers (5 min), and someone reviews and approves (5 min). **Only the last five minutes is work you would want a person doing.** Keying in supplier bills used to be the largest item at eight minutes; the system now reads the bill and runs the check in one step, so it is gone.

Why that matters: The useful thing about that list is that it is short and it is boring. Each item is a small, well-understood piece of plumbing, not a research problem — which is why the remaining two cost days rather than months. The hard part, teaching it your parts, your suppliers and your rules, is already paid for.

The last two, and what each one buys

Improvement	What stops happening	Saves	Effort
Read supplier invoices automatically	**Built. Every bill is checked as it is read**	**8 min**	**done**
Read the order mailbox directly	Nobody forwards orders any more	2 min	2–3 days
Email POs to suppliers on approval	Nobody sends the purchase orders	5 min	1–2 days
The two remaining	**14 minutes an order becomes 7**	**7 min**	**about a week**

The one worth doing first, and it is not the obvious one

Reading supplier invoices automatically. It saves the most minutes, but that is not why. Today the bill only gets checked if somebody keys it in, so the check Allied is really buying — the one catching overbilling — is the one most likely to get skipped on a busy week. Automatic reading means **every bill is checked, not just the ones someone got round to.**

Why that matters: A control that depends on somebody remembering is not a control; it is a good intention with a form attached. The value of the invoice check is not the average week when everybody keeps up. It is the quarter-end week when nobody does, which is also, reliably, the week the largest bills arrive.

Worth doing later, once you have run it a while

Improvement	What changes	Effort
Live freight rates from carriers	Real quotes instead of estimates	1–2 weeks
Live stock levels from Sage	Stock-versus-buy decisions use real numbers	1 week
Predict wear-part reorders	You call the customer before they call you	3–4 weeks

Do none of these yet

Connect Sage and Monday first, run it for a month, then decide. Automating more of a system that is not yet writing to your books adds capability nobody can use.

Why that matters: After a month, the review list tells you where the real friction is — and it is routinely not where anyone guessed. Picking the next piece from evidence costs nothing and picking it from instinct costs a fortnight.

7. Making it sturdier

Already done

These were real problems, found and fixed, each now with a check that stops them returning.

- **Proper logins.** Every action is recorded against the person who took it, from their login — not from whatever the request claimed it was.
- **Safe upgrades.** The structure of the stored records can change without losing data. Without this, the first update after go-live would quietly have broken things.
- **Exact money.** Amounts are held to the exact cent rather than approximately. In a system that matches invoices to the cent, approximately is the difference between a clean match and a phantom dispute.
- **Speed that holds up.** Reading a document used to get slower as the parts list grew. It is now the same speed whether you have 30 parts or 10,000.
- **Two people at once.** Two staff acting on the same record can no longer both succeed, which previously meant the same supplier getting two dispute letters.

The one that cannot wait

Backups

Everything the system knows lives in one file on one machine, and that file is currently copied nowhere. A nightly copy to a second drive takes about an hour to set up. **This is the only item in this guide that could lose data permanently.**

Why that matters: Every other risk here costs time. This one costs the record of what you sold, bought and certified — and in a business that holds mill certs, that record is also your evidence if a customer ever disputes a heat.

8. What it costs to run

Prices read from monday.com and sage.com on 24 October 2026. Confirm before committing.

The system adds no software fee of its own. It uses the two things you already pay for. These figures are here for the full picture, not because they are new costs.

System	Plan	Price	
Monday.com	Basic	$9 / seat / month	Enough
Monday.com	Standard	$12 / seat / month	Recommended
Monday.com	Pro	$19 / seat / month	Not needed

System	Plan	Price	
Sage 50	Pro	$132.17 / month	You have this
Sage 50	Premium	$182.50 / month	You have this
Sage 50	Quantum	$271.17 / month	You have this

You do not need the expensive Monday plan

Monday limits how often outside software may talk to it: **1,000 times a day** on the cheaper plans, 10,000 on Pro. The system uses about **two per order**. That is roughly **500 orders a day** before the cheaper plan runs out — far beyond Allied. Choosing Standard over Pro saves **$840 a year** at ten seats.

The system itself, measured

	Measured	In practice
Memory used	102 MB	Less than one browser tab. Any office PC runs it.
Storage per order	about 25 KB	Ten years of trading fits on a memory stick
Time to start	6 seconds	Restarting is not disruptive
Internet needed	No	Runs on your own network
Cost per order	$0	No usage fees of any kind

Running cost	Per year
A computer to run it on (an existing office PC works)	$0
Or a dedicated small PC, if preferred	$500–900 once
Electricity	$15–30
Backup storage	$0–70
The AI assistant, running on your own machine	$0
Total, excluding optional support	**$15–100 a year**

Why that matters: The cost that matters is not on this page. It is that somebody has to look at the review list each day. A system that flags problems nobody reads is worse than no system, because it creates the belief that the problems are being handled.

9. Keeping it running

Daily — about ten minutes

- **Open the review list.** Anything the system was unsure about is waiting, with the reason and a suggested action. This is the habit everything else depends on.
- **Deal with flagged bills.** Where a supplier overcharged, the dispute letter is already written. Read it, adjust it, send it.
- **Glance at the order feed** to confirm orders are arriving and converting.

Weekly — about half an hour

- Skim who approved what.
- Confirm last night's backup exists.
- Update any supplier prices the system flagged as old.

Monthly or quarterly

- Re-import your parts, customer and supplier lists if they have changed.
- **Restore a backup to a spare folder and check it opens.** A backup nobody has restored is a hope, not a backup.
- Look at the pattern of flags. If the same supplier appears every month, that is a commercial conversation, not a software one.

Who does what

Role	Who	What they do
Daily	Office or purchasing	Works the review list, sends disputes. No technical skill needed.
Administrator	Office manager or owner	Adds and removes staff accounts, sets what each can see, reviews the history.
Technical	Alessio, or whoever maintains it	Restarts it, applies updates, restores a backup, connects new systems.

10. If something goes wrong

What you see	Usually means	What to do
Orders stop appearing	The system stopped, or the board link was switched off	Call the technical contact

What you see	Usually means	What to do
An order is waiting, not done	A check stopped it — old price, thin margin, unknown part	Read the reason and action it
A bill looks wrongly flagged	The order in the system differs from what was agreed	Compare them; if the order is wrong, fix the order
Screens show an error	The engine stopped while the page stayed open	Call the technical contact
Numbers look wrong after an update	A change went in without rehearsal	Restore last night's backup and report it

What it will never do

- Send anything to a customer or supplier without a person approving it.
- Approve a supplier bill that does not match an order you raised.
- Certify material it cannot verify.
- Send any Allied information outside the building.
- Change something in Sage that a person has already posted.

Allied Universal Operations Bridge — How It Works — 24 October 2026. The system currently passes 150 automatic checks, 60 full order run-throughs and 27 deliberate attempts to break it. Sage and Monday connections are built and tested but not yet switched on.